Hassan Mehmood

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LinkedIn | GitHub | Personal Website | Tableau Profile

Summary

Data Scientist / ML Engineer / AI Engineer with 3+ years of experience delivering production ML systems, automated ETL pipelines, and BI solutions in fintech and B2B environments. Skilled in Python, SQL, TensorFlow, PyTorch, Airflow, and Power BI, with a focus on scalable ML deployment, data quality, and actionable insights. Passionate about explainable AI, AI Security, and Privacy, ensuring responsible and trustworthy ML solutions.

Skills

ML & Data: Python, SQL, TensorFlow, PyTorch, Scikit-Learn, Statistical Modeling, Time Series Analysis, Predictive Analytics, JavaScript, React

Engineering & Ops: Docker, CI/CD, Git, REST APIs, Data Pipeline Orchestration, Model Monitoring & Versioning, Lang-chain, Lang-graph, Vector Databases, RAG Systems, MLFlow, DVC, MLOps

BI & Analytics: Power BI, Tableau, ETL Pipelines, Data Visualization, A/B Testing

Experience

Founding Data Scientist, Book Depot Inc. – Thorold, ON

July 2026 - Present

- Engineered and deployed an automated dynamic pricing engine using Causal DML to optimize price elasticity, sales velocity, and inventory signals across a 60K+ SKU catalog.
- Scaled dynamic pricing strategies across Direct-to-Consumer (DTC), third-party marketplaces, and wholesale channels, driving an average revenue increase of over 5%.
- Built advanced behavioral clustering engines using unsupervised learning to segment customer demand patterns and enhance catalog visibility.
- Developed inventory pacing and demand forecasting models that actively optimize multi-channel pricing to improve slow-moving inventory sell-through.
- Engineered scalable ML pipelines, achieving 15x faster processing through parallelized computation.

Data Scientist, Iteration Matrix – Toronto, ON

Sep 2025 – July 2026

- Architected and productionized ML systems for a centralized Supplier Mastering module, ROI attribution, and audit recovery, enabling scalable, data-driven decisions across business units.
- Reduced reporting errors from 25 % to less than 5% through large-scale data preprocessing, feature engineering, and robust data pipelines on structured and unstructured datasets.
- Owned end-to-end ML-Ops and model lifecycle, including CI/CD, containerized deployments, automated retraining, drift detection, and real-time monitoring.

Data Science Analyst Intern, EQ Bank – Toronto, ON

May 2025 – Aug 2025

Model Performance Monitoring System:

- Built and automated end-to-end ETL pipelines for model performance data, enabling real-time Power BI dashboards and reducing manual reporting by 90%.
- Designed a normalized relational database schema to store model KPIs for multi-period analysis, supporting historical trend tracking and decision-making.

Enterprise Data Quality Framework:

- Spearheaded development of a high-performance Data Quality Management System (DQMS) in Python to continuously profile and validate over 1 TB mission-critical SQL Server database.

Instructional Specialist / Tutor, University of Toronto – Toronto, ON

Sep 2022 – Aug 2024

- Mentored 50+ students in data analytics, machine learning, and time series analysis.
- Developed learning materials in Python, SQL, and TensorFlow, improving student engagement.
- Simplified complex AI and blockchain concepts, consistently receiving positive feedback for clarity.

Education

University of Ontario, Institute of Technology – Oshawa, ON

Masters, Business Analytics and Artificial Intelligence, 2025

GPA: 3.99/4.3

- Business Analytics, Programming and Data Processing, Data Visualizations, Ethics in AI.
- Digital Transformation, AI Programming, Big Data System Design, AI Security Privacy and Trust.

George Brown College – Toronto, ON

Postgraduate Program, A.I. Solutions Development Program, 2024

GPA: 3.76/4.0

- Big Data, Machine Learning, Deep Learning, Data Visualization, Data Management.

University Of Toronto – Toronto, ON

Certificate, Financial Technology, 2022

University Of Health Sciences – Pakistan

Bachelors, Health Sciences, 2015

Projects

Macro-Driven Equity Positioning System

Python · Lang Chain · Ollama (Q-wen 2.5 3B) · Weaviate · FRED API · Pydantic

- Engineered a three-stage RAG pipeline combining semantic query planning, vector retrieval, and structured LLM classification to generate macro regime outputs with confidence scores, reasoning, and historical analogues.
- Built a rule-based macro engine deriving five core signals (growth, inflation, liquidity, risk, labour) with confirmation layers and persistence logic to reduce noise and improve regime stability.
- Designed a mapping layer translating 8 macro regimes into actionable equity strategies, including sector tilts, factor exposures, and stock selection archetypes.
- Deployed the inference pipeline using Ollama, with optimized embeddings and vector storage in Weaviate for efficient, low-latency retrieval from economic research corpora.

Certificates

- IBM, Full Stack Software Developer Certificate
- IBM, Professional Data Engineering Certificate
- Microsoft, Azure Data Scientist Associate
- EDHEC, Investment Management with Python and Machine Learning

Awards

- Deans Honor List, George Brown College – Fall 2023
- Deans Honor List, George Brown College – Winter 2024